

Intro to Git

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Introduction

- Git is a version control system used to track versions of files.
- It is a critical skill in scientific programming and collaborative research.
- Git can be used locally to track changes in a directory. There are stored in a folder called `.git` that will exist in the directory once git has been initialised
- Using git locally can be useful for keeping track of changes in solo projects, however most developers will host their repository online also
- This has the obvious advantage of working as an online backup, but also allows multiple developers to work on the same codebase
- GitHub and GitLab are two of the most common platforms for hosting git repositories
- The platform we will be using Github: <https://github.com/>

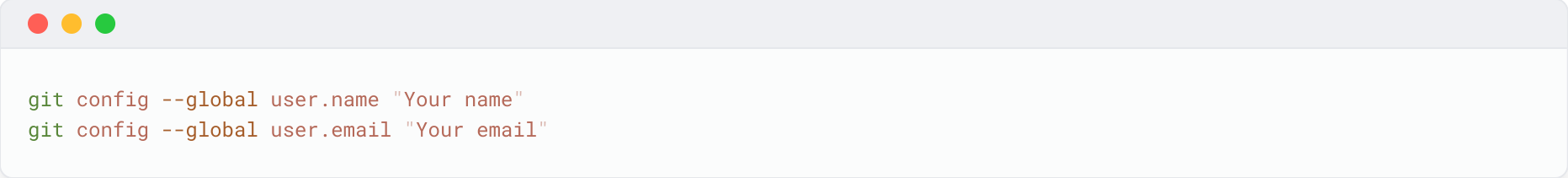
Creating a Github account

- Creating a Github account is very straightforward.
- All you have to do is to click “Sign up” in the top right of <https://github.com/>.
- You will lose access to your @ucdconnect.ie email when you graduate, so I would recommend using a different email address

Adding an SSH key

- The first step is to add our SSH key to Github.
- Settings-> SSH and GPG keys.
 - Linux: <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account?platform=linux>
 - Mac: <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account?platform=mac>
 - Windows: <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account?platform=windows>

Final Setup



```
git config --global user.name "Your name"  
git config --global user.email "Your email"
```

This should also be done on any remote machines you are using to make changes to your code

Useful git commands

- `git init`: initialises a git repository within a directory
- `git status`: gives a summary of the status of the repo, including the current branch, any untracked files, and any tracked files that have been changed
- `git add 'filename'`: used to start tracking changes to a file, or to update a tracked file which has been changed
- `git commit`: commits changes, which creates a snapshot of all tracked files in the repo at the moment of the commit. Usually used as `'git commit -m "Commit message"'` with a summary of the updates for that snapshot
- `git clone`: create a local copy of a remote repository
- `git pull`: update a local copy of a repo with changes to the remote version
- `git push`: update a remote repo with your local changes
- `git log`: Give a list of previous commits on your branch

Practical

1/11



STEP 1

```
mkdir practical_example_dir  
cd practical_example_dir  
git init  
git status
```



STEP 2

```
touch sample.txt  
git status
```



STEP 3

```
echo "This is content" >> sample.txt  
git add sample.txt  
git status
```



STEP 4

```
git commit -m "Added sample.txt"
git status
```



STEP 5

```
git push
```

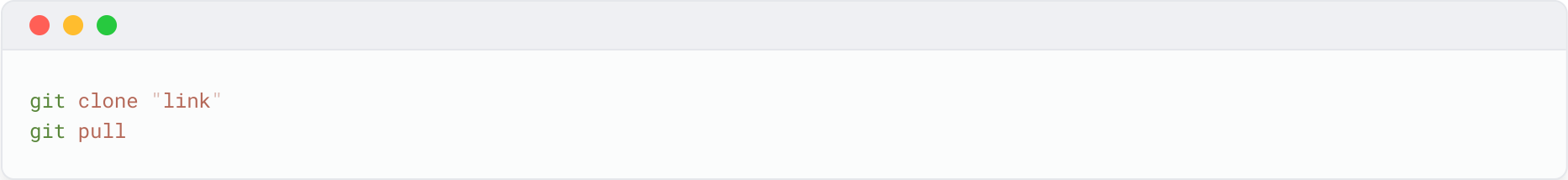


STEP 6

```
git remote add origin "your link here"
git branch -M main
git push -u origin main
```

Accessing Changes

- We have now moved our changes into the remote repo.
- How do we access these on another machine?



```
git clone "link"  
git pull
```

Branches

- Different versions of a repo that can be edited independently of each other.
- Changes can be incorporated into the `main / master` branch.



```
git branch new_branch  
git checkout new_branch
```

- How do you get the changes from one branch into another?

Merging

- The process of combining branches is called merging.
- To get the changes from `hotfix` into `main`:



```
git checkout main  
git merge hotfix
```

- Pushing to a branch will create a merge/pull request. Here you lay out your changes and why you made them. This request must then be approved before `git merge` can be run.